

AI + Quantum AI Senior Engineer Learning Plan (v3)

Senior-Level Full-Stack AI + Quantum AI — Learning Plan (v3, mastery-sequenced)

Owner: MD Hashim · **No fixed deadline** · **Total effort:** ~1,100 focused hours · **Sequencing principle:** bottom-up mastery

What changed and why (read this first)

The August quantum project is off the table, and you asked for the most *suitable learning pattern* with the *optimal time each topic actually needs* — no rigid deadline. That flips the design philosophy entirely.

- **v2 was deadline-driven** — it compressed content into ~29 weeks and front-loaded quantum for August. Every phase was fighting a clock.
- **v3 is mastery-driven** — it sequences everything **bottom-up**: fix the mathematical foundation first (your stated gap), then add each domain only once its prerequisites are genuinely in hand. Nothing is rushed; nothing is out of order.

The sequencing logic:

1. **Foundations come first, properly — not as a sprint.** Your self-diagnosed gap is math and the *why* beneath techniques. Everything downstream — segmentation architectures, transformer theory, VQE, QML — rests on linear algebra, calculus, and probability. This is the single highest-leverage phase, so it gets real time.
2. **Your active domain (CV / medical segmentation) comes second.** Here's the key insight: the foundations phase includes Karpathy's *Zero to Hero*, which builds attention/transformers **from scratch**. So by the end of Phase 1 you already understand attention mechanically — which means your segmentation work (both CNN- and transformer-based) can come right after foundations instead of waiting behind the full deep-learning tour. You get early alignment with your real job *and* correct prerequisite order.
3. **Deep-learning theory then goes broad and deep** — the advanced transformer variants, self-supervised learning, and generative/diffusion theory, plus the training-dynamics math (optimization, information theory). This generalizes and deepens what you used in Phase 2.
4. **Quantum gets a proper, unhurried block** — no longer squeezed for a deadline. And because you'll have done the ML foundations, QML (which is really kernel methods + variational optimization) lands far deeper.
5. **Production breadth, then a specialization + integrative capstone** — the culmination draws on everything.

Math is no longer a parallel "sprint." The core (linear algebra, calculus, probability) is a real first phase; the specialized pieces (optimization, information theory, group theory) are threaded into the exact phase that uses them.

Placement note on segmentation: I put it at Phase 2 so it directly follows the foundation it depends on and stays close to your day-to-day work. If you'd rather push it later and do the full DL-theory phase first, it's a clean swap — but Phase 2 is the better default now that Karpathy's series gives you attention up front.

Pacing (there is no deadline — the hours are the target, the calendar is yours)

The number that matters is **effort (~1,100 hours)**, not a date. As a *soft* planning aid: at a sustainable **~25-30 hrs/week** the whole plan is roughly **9-11 months** of elapsed time; at a lighter ~15 hrs/week it's closer to 18 months. Pace it to your energy and life. Two principles that matter more than speed:

- **Depth over pace.** A phase is "done" when you pass its whiteboard self-test cold, not when a week ends. If linear algebra takes you three extra weeks, take them — the compounding return is enormous.
 - **Protect sustainability.** This is a marathon layered on a full-time senior role. Keep at least one genuine rest block per week. Consistency at 20 hrs/week beats heroics at 40 that collapse in month three.
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Phase summary (the whole plan at a glance)

Phase	Focus	Effort	What it unlocks
1	Foundations — mathematics + neural nets from first principles	~190 hrs	The bedrock for everything; includes attention from scratch
2	Applied CV & Medical Image Segmentation (your active domain)	~185 hrs	Senior depth in your day-to-day work, now grounded
3	Deep Learning theory & modern architectures	~150 hrs	The broad/deep DL backbone; generative + training theory
4	Quantum Computing & QML	~275 hrs	Rigorous theory beneath your existing VQE/ADAPT/DMET work
5	Full-stack / production AI systems	~160 hrs	Production breadth + systems-design interview readiness
6	Specialization + integration + interview prep	~135 hrs	Ties imaging, quantum, and drug-discovery together

PHASE 1 — Foundations: Mathematics + Neural Networks from First Principles

~190 hrs · the bedrock — do not rush this

Goal: Build the mathematical maturity your senior transition depends on, and turn it into working neural networks so backprop and "the why" are bone-deep. By the end you can re-derive the core results on paper *and* you understand attention mechanically (via Karpathy's GPT build), which unlocks Phase 2.

1.1 Linear algebra (~68 hrs) — the highest-leverage topic in the entire plan

Underpins both deep learning and quantum. Do it thoroughly.

- **3B1B — Essence of Linear Algebra** (geometric intuition first; watch twice if needed) — ~8 hrs https://www.youtube.com/playlist?list=PLZHQObOWTQDPD3MizzM2xVFitgF8hE_ab
- **MIT 18.06 — Gilbert Strang** (L1, 3, 5, 6, 9, 10, 14, 15, 16, 21, 22, 25, 29, 30 + problems) — ~35 hrs https://ocw.mit.edu/courses/18-06-linear-algebra-spring-2010/video_galleries/video-lectures/
- **MIT 18.065 — Matrix Methods (Strang)** — the *applied* view: SVD, PCA, low-rank, gradient descent through a linear-algebra lens; closest to ML reality — ~25 hrs <https://ocw.mit.edu/courses/18-065-matrix-methods-in-data-analysis-sign-al-processing-and-machine-learning-spring-2018/>
- Reference kept open: **Matrix Cookbook** — <https://www.math.uwaterloo.ca/~hwolkowi/matrixcookbook.pdf>

Whiteboard checkpoint: derive SVD from eigendecomposition; explain geometrically what a rank-r approximation does; show $A^T A$ is PSD; derive backprop through a linear layer with matrix calculus.

1.2 Calculus + matrix calculus (~22 hrs)

- **3B1B — Essence of Calculus** — ~8 hrs <https://www.youtube.com/playlist?list=PLZHQObOWTQDMsr9K-rj53DwVRMYO3t5Yr>
- **Matrix Calculus for Deep Learning (Parr & Howard)** — best short article on Jacobians for ML; plus Matrix Cookbook practice — ~14 hrs <https://explained.ai/matrix-calculus/>

1.3 Probability & statistics (~45 hrs)

Underpins loss functions, generative models, uncertainty, and QML. Do a serious pass.

- **Harvard Stat 110 — Joe Blitzstein** (the best intro-probability course; work through conditional expectation, MGFs, key distributions) — ~45 hrs <https://projects.iq.harvard.edu/stat110/youtube>
- Alternative if Blitzstein feels slow: **MIT 6.041 / RES.6-012 — Tsitsiklis** <https://ocw.mit.edu/courses/res-6-012-introduction-to-probability-spring-2018/>

Whiteboard checkpoint: derive MLE for a Gaussian; explain conditional expectation as a projection; state and use

Bayes cleanly on a non-trivial example.

1.4 Neural networks from first principles (~55 hrs) — the bridge from math to DL

Non-negotiable even with your experience. Do all 10 videos **with the exercises**, re-implementing each without looking at his code. This is where "the why" lives, and the GPT build gives you attention mechanically — which Phase 2 relies on.

- **Andrej Karpathy — Neural Networks: Zero to Hero** — ~55 hrs Playlist: <https://www.youtube.com/playlist?list=PLAqhlrjxkxW123v9cThsA9GvCAUhRvKZ> · Hub: <https://karpathy.ai/zero-to-hero.html> Order: micrograd → makemore 1-5 (incl. BatchNorm internals, manual backprop, WaveNet) → GPT from scratch → tokenizer → reproduce GPT-2.

Phase 1 self-test: hand-derive backprop for a 2-layer MLP; implement micrograd-style autodiff from memory; explain why BatchNorm helps and what breaks without it; whiteboard scaled dot-product attention (you built it — now derive the $\sqrt{d_k}$ scaling).

PHASE 2 — Applied Computer Vision & Medical Image Segmentation

~185 hrs · your active domain — now with full grounding · **PRIORITY: high (your day-to-day work)**

Goal: Own MRI / fluorescence brain segmentation end-to-end — physics → preprocessing → architectures → losses → 3D → uncertainty → evaluation. You arrive here already understanding attention and backprop, so nnU-Net, UNETR, and the math beneath the loss functions land hard. By the end you can whiteboard nnU-Net's design choices cold and defend each in a project discussion.

2.1 Imaging physics (~30 hrs) — most engineers skip this and it shows in their models

- **Allen Elster — Questions & Answers in MRI** (the reference you'll return to) — ~8 hrs <https://mriquestions.com/index.html>
- **OHBM Educational — MRI physics** playlist — ~8 hrs <https://www.youtube.com/@ohbmonline/playlists>
- **Paul Callaghan — "Magnetic Resonance" (Otago)**, first 4 lectures — ~6 hrs https://www.youtube.com/playlist?list=PLqEMVgX2VgZcG_xR1QRdsuc-X_zEgMx-Z
- **iBiology — fluorescence microscopy** primer — ~8 hrs <https://www.youtube.com/playlist?list=PLF513KEDjY9YBNhwMfX6jMI3PlpdW9TMP>

Deliverable: 1-page note — T1 vs T2 vs FLAIR vs T1CE (tissue appearance, why each exists, artifacts to preprocess: motion, bias field, partial volume); for fluorescence: photobleaching, channel bleed-through, autofluorescence and their segmentation impact.

2.2 CNNs + convolution math + U-Net family + losses (~50 hrs)

- **CS231n 2017 — Justin Johnson** (CNN theory: L5, 9, 11 + notes) — ~18 hrs <https://www.youtube.com/playlist?list=P3LFW7Lu3i5jvHM8ljYj-zLqRF3EO8sYv> · <https://cs231n.github.io/>
- **Convolution arithmetic (Dumoulin & Visin)** — canonical reference for shapes/strides/padding/dilation (2D and 3D) — ~5 hrs <https://arxiv.org/abs/1603.07285> · https://github.com/vdumoulin/conv_arithmetic
- **U-Net family, in order, deriving output sizes by hand** — ~15 hrs U-Net: <https://arxiv.org/abs/1505.04597> · V-Net (3D): <https://arxiv.org/abs/1606.04797> · Attention U-Net: <https://arxiv.org/abs/1804.03999> · **nnU-Net (read twice + supplementary):** <https://www.nature.com/articles/s41592-020-01008-z>
- **Loss functions for medical segmentation** (your 95/5 imbalance) — survey + key losses, then **from-scratch PyTorch implementation** (Dice, GDL, Focal, Tversky, Focal Tversky, Boundary, compound DiceCE; unit-test on synthetic imbalanced data; plot gradients vs class ratio) — ~12 hrs Survey: <https://arxiv.org/abs/2006.14822> · Generalized Dice: <https://arxiv.org/abs/1707.03237> · Focal Tversky: <https://arxiv.org/abs/1810.07842> · Boundary loss: <https://arxiv.org/abs/1812.07032>

Whiteboard checkpoint: draw U-Net from memory with tensor shapes; explain why skip connections work; derive the Dice gradient and its instability on empty masks; explain why nnU-Net's "no new architecture" thesis is profound.

2.3 nnU-Net + MONAI + microscopy-native hands-on (~45 hrs) — the biggest applied block

- **MONAI tutorials** (clone, run, modify the brain-seg notebooks) — ~15 hrs <https://github.com/Project-MONAI/tutorials> · Videos: <https://www.youtube.com/@projectmonai/videos>

- **nnU-Net v2** — read the *source*; it teaches more than the paper — ~8 hrs <https://github.com/MIC-DKFZ/nnUNet> · MONAI integration: <https://github.com/Project-MONAI/tutorials/blob/main/nnunet/README.md> · Kitware comparison: <https://www.kitware.com/developing-custom-3d-medical-image-segmentation-solutions-using-out-of-the-box-pipelines-in-monai/>
- **DigitalSreeni (Python for Microscopists)** — the most microscopy-native channel there is; closer to your fluorescence + volumetric work than any Stanford course. ~6–8 videos, code-along — *~14 hrs* Channel: <https://www.youtube.com/@DigitalSreeni/playlists> · Code: https://github.com/bnsreenu/python_for_microscopists · U-Net playlist: <https://www.youtube.com/playlist?list=PLZsOBAYNTZwbR08R959iCvYT3qzhxvGOE>
- **AI for Medical Diagnosis (Rajpurkar, deeplearning.ai)** — audit free; the brain-MRI U-Net segmentation + evaluation modules are on-point — ~8 hrs <https://www.coursera.org/learn/ai-for-medical-diagnosis>

Deliverable: reproduce nnU-Net on a public brain dataset (BraTS subset or a Medical Segmentation Decathlon task); write up which defaults it chose, *why*, and what you'd change for fluorescence microscopy.

2.4 Transformer-based segmentation + foundation models (~20 hrs)

You understand attention from Phase 1, so these land as applications, not mysteries.

- ViT (re-derive attention if rusty): <https://arxiv.org/abs/2010.11929> — ~3 hrs
- TransUNet / Swin-UNet / UNETR / Swin UNETR — ~9 hrs <https://arxiv.org/abs/2102.04306> · <https://arxiv.org/abs/2105.0537> · <https://arxiv.org/abs/2103.10504> · <https://arxiv.org/abs/2201.01266>
- SAM / MedSAM: <https://arxiv.org/abs/2304.02643> · <https://www.nature.com/articles/s41467-024-44824-z> — ~5 hrs
- MedNeXt (a ConvNet that beats transformers on many medical tasks — keeps you honest): <https://arxiv.org/abs/2303.09975> — ~3 hrs

Whiteboard checkpoint: compare inductive biases of CNN vs ViT vs hybrid — when each wins; role of foundation models in low-data medical regimes; SAM's promptable design and what MedSAM changes.

2.5 3D, uncertainty, evaluation + capstone (~40 hrs)

- **TorchIO** (3D augmentation done right) — ~10 hrs <https://torchio.readthedocs.io/> · <https://www.youtube.com/playlist?list=PLfXlfDgnRPfIEHFx2N6mIbn-MGFOPiX2>
- **Uncertainty (Yarin Gal — MC Dropout, deep ensembles, evidential)** — ~5 hrs https://www.youtube.com/results?search_query=yarin+gal+uncertainty+deep+learning
- **Metrics Reloaded** (definitive 2024 paper on choosing eval metrics) — ~4 hrs <https://www.nature.com/articles/s41592-023-02151-z>
- **Domain shift in medical imaging (Stanford AIMI)** — ~4 hrs <https://aimi.stanford.edu/education/educational-resources>
- **3D-conv whiteboard exercise:** input (D=64, H=128, W=128, C=4), kernel (3,3,3), stride (1,2,2), padding (1,1,1), 32 out-channels — compute output shape AND param count by hand; repeat for transposed + dilated conv — ~2 hrs
- **PHASE 2 CAPSTONE (~15 hrs)** — notebook + 3-page write-up on your mouse-brain data (or a BraTS subset): (1) preprocessing with physics justification, (2) architecture ablation across 3 architectures, (3) loss-function ablation, (4) uncertainty quantification + failure-mode analysis. The senior-engineer artifact — and exactly what a panel wants to see.

Phase 2 self-test: whiteboard U-Net forward pass + skips with shapes; explain why nnU-Net usually beats fancier architectures (and when it doesn't); for a new dataset, walk through diagnosing bias-field artifacts, class imbalance, label noise, domain shift.

PHASE 3 — Deep Learning Theory & Modern Architectures

~150 hrs · the broad, deep DL backbone

Goal: Generalize beyond what Phase 1–2 taught you: the advanced transformer variants, self-supervised learning, generative/diffusion theory, and the training-dynamics math (optimization + information theory). By the end you can defend architecture and training choices on theoretical grounds.

3.1 Optimization for deep learning (~28 hrs) — threaded here because this is where training theory lives

- **Sebastian Ruder — gradient-descent optimization overview** — ~3 hrs <https://www.ruder.io/optimizing-gradient-descent/>
- **Stanford EE364A — Convex Optimization (Boyd)** — selected lectures for the convex intuitions that even non-convex DL builds on — ~22 hrs <https://www.youtube.com/playlist?list=PL3940DD956CDF0622> · book: <https://web.stanford.edu/~boyd/cvxbook/>
- Optimizer/training-dynamics deep dive (SGD, momentum, Adam, warmup, schedules, second-order intuitions) — ~3 hrs

3.2 Information theory for ML (~18 hrs) — the language of loss functions

- **David MacKay — Information Theory, Inference, and Learning Algorithms, ch 1-6** (entropy, KL, mutual info — underpins cross-entropy, VAEs, diffusion, contrastive learning) — ~18 hrs <https://www.inference.org.uk/itila/book.html>

Whiteboard checkpoint: derive cross-entropy from KL; show NLL minimization = MLE; derive the ELBO; explain mutual information and where it appears in InfoNCE / contrastive losses.

3.3 Transformers in depth + modern variants (~40 hrs)

- **Stanford CS25 — Transformers United** (6-8 talks) — <https://web.stanford.edu/class/cs25/> · https://www.youtube.com/playlist?list=PLoROMvodv4rNijRchCzutFw5ItR_Z27CM
- **Lilian Weng's blog** (attention, transformer, RLHF) — <https://lilianweng.github.io/>
- **The Annotated Transformer** (code + math line by line) — <http://nlp.seas.harvard.edu/annotated-transformer/>
- **Modern tricks** (~30 min each): RoPE (<https://arxiv.org/abs/2104.09864>) · RMSNorm (<https://arxiv.org/abs/1910.07467>) · FlashAttention (<https://arxiv.org/abs/2205.14135>) · GQA/MQA (<https://arxiv.org/abs/2305.13245>) · SwiGLU (<https://arxiv.org/abs/2002.05202>) · MoE / Switch (<https://arxiv.org/abs/2101.03961>)

Whiteboard checkpoint: explain RoPE geometrically; why pre-LN beats post-LN; why LayerNorm not BatchNorm in transformers; compute a transformer block's param count from `d_model`, `n_heads`, FFN ratio.

3.4 Vision transformers + modern ConvNets (~25 hrs)

- ViT lectures (Lucas Beyer) — https://www.youtube.com/results?search_query=lucas+beyer+vision+transformer
- ConvNeXt (<https://arxiv.org/abs/2201.03545>) · DINOv2 (self-supervised, relevant to label-scarce medical work — <https://arxiv.org/abs/2304.07193>) · MAE (<https://arxiv.org/abs/2111.06377>)
- **Hugging Face Computer Vision course** — <https://huggingface.co/learn/computer-vision-course>

3.5 Generative modeling & diffusion (~38 hrs)

Diffusion math underpins the modern field and shows up in medical image synthesis/augmentation.

- **Stanford CS236 — Deep Generative Models (Ermon)** (L1-3, 6-8, 11-13) — <https://www.youtube.com/playlist?list=PLoROMvodv4rPOWA-omMM6STXaWW4FvJT8>
- DDPM (<https://arxiv.org/abs/2006.11239>) · Yang Song, score-based unification + blog (<https://yang-song.net/blog/2021/score/>) · Lilian Weng diffusion post (<https://lilianweng.github.io/posts/2021-07-11-diffusion-models/>)
- **Hugging Face Diffusion course** (units 1-2 hands-on) — <https://huggingface.co/learn/diffusion-course>

Phase 3 self-test: derive the diffusion loss (variational + score-matching views); compare positional encodings (sinusoidal/learned/RoPE/ALiBi); explain the role of self-supervised pretraining in low-data regimes.

PHASE 4 — Quantum Computing & Quantum Machine Learning

~275 hrs · rigorous theory beneath your existing hands-on work — now unhurried

Goal: Match your production VQE/ADAPT-VQE/DMET-VQE experience with real theory. Derive VQE from the variational principle, explain barren plateaus mathematically, defend ansatz choices, and discuss QML's genuine limits honestly (per Schuld's critiques). This is *depth on top of experience* — and because you've now done the ML foundations, QML (really kernel methods + variational optimization) lands far deeper.

4.1 Group theory & symmetry (~22 hrs) — threaded here; also pays off in Phase 6 geometric DL

- **Group theory for physicists** (pick one lecture series) — ~15 hrs https://www.youtube.com/results?search_query=group+theory+for+physicists+lectures
- **Geometric Deep Learning lectures (Bronstein)** — the Lie-group / equivariance angle — ~7 hrs <https://geometricdeeplearning.com/lectures/>

4.2 Quantum-mechanics math: Hilbert spaces, operators, tensor products (~25 hrs)

- **MIT 8.04 — Quantum Physics I (Adams)** (L6-12 if the experimental motivation feels slow) — ~15 hrs https://ocw.mit.edu/courses/8-04-quantum-physics-i-spring-2013/video_galleries/lecture-videos/
- **Quantum Country** (spaced-repetition intro — fast and sticky) — ~6 hrs <https://quantum.country/qcvc>
- (Optional, rigor-minded) Frederic Schuller — Geometric Anatomy of Theoretical Physics — ~4 hrs sampled https://www.youtube.com/playlist?list=PLPH7f_7ZlzxTi6kS4vCmv4ZKm9u8g5yic

Whiteboard checkpoint: define a Hilbert space rigorously; tensor-product two qubit states; explain unitarity, Hermitian operators, why measurement outcomes are eigenvalues; distinguish pure vs mixed states + the density matrix.

4.3 Quantum computing fundamentals: the Watrous / IBM track (~80 hrs)

The best free quantum-information series that exists — John Watrous's four IBM courses.

- **Basics of Quantum Information** — ~25 hrs <https://learning.quantum.ibm.com/course/basics-of-quantum-information> · YouTube: <https://www.youtube.com/playlist?list=PLOFEBzvs-VvrXTMy5Y2lqmSaUjfnhvBHR>
- **Fundamentals of Quantum Algorithms** — ~25 hrs <https://learning.quantum.ibm.com/course/fundamentals-of-quantum-algorithms>
- **General Formulation of Quantum Information** (density matrices, channels, POVMs — matters for noise/error mitigation) — ~18 hrs <https://learning.quantum.ibm.com/course/general-formulation-of-quantum-information>
- **Foundations of Quantum Error Correction** — ~12 hrs <https://learning.quantum.ibm.com/course/foundations-of-quantum-error-correction>
- Companion (free): **Nielsen & Chuang**, ch 1-4, 8, 10 (read alongside)

4.4 Variational quantum algorithms — VQE, ADAPT-VQE, DMET, QAOA (~45 hrs)

You've shipped this; now own the math.

- **PennyLane Codebook — Variational Quantum Algorithms** (work the exercises) — <https://pennylane.ai/codebook>
- **PennyLane quantum-chemistry demos** (run *all* the derivations behind the code you've used) — https://pennylane.ai/qml/demos_quantum-chemistry
- Papers, re-derived: VQE (<https://www.nature.com/articles/ncomms5213>) · ADAPT-VQE (<https://www.nature.com/articles/s41467-019-10988-2>) · DMET-VQE / fragment methods (<https://arxiv.org/abs/2108.08987>) · QAOA (<https://arxiv.org/abs/1411.4028>) + Hadfield review (<https://arxiv.org/abs/1709.03489>) · Pauli decomposition + measurement grouping (shot-budget engineering, <https://arxiv.org/abs/1907.13117>)

Whiteboard checkpoint: derive the variational principle; walk VQE end-to-end (ansatz choice, parameter-shift rule, optimizer); explain barren plateaus mathematically + mitigations; why ADAPT-VQE is more parameter-efficient; when DMET fragmentation is worth the overhead.

4.5 Quantum machine learning theory (~42 hrs)

- **PennyLane QML topic page (Schuld-curated)** — the most honest overview in the field — <https://pennylane.ai/topics/quantum-machine-learning>
- **Maria Schuld — "Taking stock of QML: a critical perspective"** — the talk every senior should watch (tells you what's hype vs real) — https://www.youtube.com/results?search_query=maria+schuld+taking+stock+quantum+machine+learning
- Schuld & Petruccione, *Machine Learning with Quantum Computers* — ch 4-7 (kernels, variational, QNNs)
- Papers: "QML models are kernel methods" (<https://arxiv.org/abs/2101.11020>) · Barren plateaus (<https://www.nature.com/articles/s41467-018-07090-4>) · Expressibility & entangling capability (<https://arxiv.org/abs/1905.10876>) · "Is quantum advantage the right goal?" (<https://arxiv.org/abs/2203.01340>) · PQC encodings (<https://arxiv.org/abs/2008.08605>)
- Structured courses to pull lectures from: Christophe Péré (<https://github.com/Christophe-pere/QML-Course>) · Morten Hjorth-Jensen, Oslo (<https://github.com/CompPhysics/QuantumComputingMachineLearning>)

4.6 Quantum chemistry on quantum computers + capstone (~60 hrs)

- **Qiskit Global Summer School — recordings** (chemistry + variational tracks) — <https://www.youtube.com/@qiskit/playlists> — ~15 hrs
- **IBM Quantum Learning platform + Sample-based / Krylov Quantum Diagonalization (QDA)** (IBM's newest near-term technique) — <https://learning.quantum.ibm.com/> · <https://www.ibm.com/quantum/blog/iql-migration> — ~15 hrs
- **PHASE 4 CAPSTONE** (~30 hrs) — take one molecule from your Lilly work (keto-enol tautomerism or HATU coupling). Re-implement three ways — vanilla VQE + UCCSD, ADAPT-VQE, DMET-VQE — benchmark against classical CCSD/6-31G. The 4-page write-up defends *every* choice on theoretical grounds (ansatz, optimizer, measurement grouping, noise mitigation, basis set) and includes a barren-plateau analysis.

Phase 4 self-test: derive VQE from the variational principle; explain the parameter-shift rule and why it works; whiteboard a barren-plateau argument; VQE vs QAOA — when each, and when *no* quantum algorithm; honest take on where QML is overhyped vs promising (cite Schuld); walk through DMET fragmentation and the classical-quantum interface.

PHASE 5 — Full-Stack / Production AI Systems

~160 hrs · production breadth + systems-design readiness

Goal: Upgrade from "I built it" to "I can defend every architectural choice and design the next-gen version" across LLM systems, RAG, MLOps, distributed training, and serving.

5.1 LLM internals (~32 hrs)

- **Stanford CS336 — Language Modeling from Scratch** (currently the best LLM-systems course) — <https://stanford-cs336.github.io/spring2024/> · https://www.youtube.com/playlist?list=PLoROMvodv4rOY23Y0BoGoBGgQ1zmU_MT_
- **Hugging Face NLP course** (units new to you) — <https://huggingface.co/learn/nlp-course>

5.2 RAG at senior level (~28 hrs)

- **Hamel Husain — "What we learned from a year of building with LLMs"** (Parts I-II) — <https://www.oreilly.com/radar/what-we-learned-from-a-year-of-building-with-llms-part-i/>
- **Eugene Yan — LLM patterns** — <https://eugeneyan.com/writing/llm-patterns/>
- **Anthropic — contextual retrieval** — <https://www.anthropic.com/news/contextual-retrieval>
- Reranking, hybrid search, query rewriting, evals; **LlamaIndex** (<https://docs.llamaindex.ai/>) · **RAGAS** (<https://docs.ragas.io/>) · **DSPy** (<https://dspy.ai/>)

5.3 MLOps + ML systems design (~42 hrs)

- **Chip Huyen — Designing Machine Learning Systems** / interviews book — <https://huyenchip.com/ml-interviews-book/contents/8.1.1-ml-systems-design.html>
- **Stanford CS329S — ML Systems Design** — <https://stanford-cs329s.github.io/>
- **Made With ML** (hands-on) — <https://madewithml.com/> · **Full Stack Deep Learning** — <https://fullstackdeeplearning.com/course/2022/>

5.4 Distributed training + inference optimization (~25 hrs)

- **Sebastian Raschka** (distributed training + efficient fine-tuning) — <https://magazine.sebastianraschka.com/>
- **HF — efficient training on multiple GPUs** — https://huggingface.co/docs/transformers/en/perf_train_gpu_many
- **vLLM** (paper + run it) — <https://arxiv.org/abs/2309.06180> · quantization (GPTQ/AWQ/GGUF/FP8) · **ZeRO** — <https://arxiv.org/abs/1910.02054>

5.5 Eval / observability / safety + systems-design interview prep (~33 hrs)

- **Anthropic — Constitutional AI** — <https://www.anthropic.com/news/constitutional-ai> · **Eugene Yan — LLM evals** — <https://eugeneyan.com/writing/llm-evaluators/>
- Observability tool (LangSmith / Phoenix / Helicone) — run one on a real RAG system.
- **Systems-design interview prep** — Chip Huyen's book (<https://huyenchip.com/ml-interviews-book/>); practice on a whiteboard, recorded, self-critiqued: "RAG assistant for HCPs, 10K queries/day on 50K docs"; "brain-seg pipeline ingesting new-institution data weekly with domain-shift handling"; "retraining + monitoring for a drug-discovery digital twin."

PHASE 6 — Specialization + Integration + Interview Prep

~135 hrs · the culmination — ties imaging, quantum, and drug-discovery together

Goal: Close the geometric-deep-learning / molecular-AI gap that connects all your work, then integrate everything and become interview-ready as a senior engineer.

6.1 Geometric deep learning + molecular ML (~45 hrs)

- **Bronstein et al. — Geometric Deep Learning course** — <https://geometricdeeplearning.com/lectures/> · proto-book: <https://arxiv.org/abs/2104.13478> · Veličković GNN lectures: https://www.youtube.com/results?search_query=petar+velickovic+gnn+lectures
- **Molecular ML** — DeepChem (<https://deepchem.io/tutorials/>) · Therapeutics Data Commons (<https://tdcommons.ai/>) · SchNet (<https://arxiv.org/abs/1706.08566>) · DimeNet++ (<https://arxiv.org/abs/2011.14115>) · MolFormer (<https://arxiv.org/abs/2106.09553>) · equivariant nets, Cohen & Welling (<https://arxiv.org/abs/1602.07576>)

6.2 Protein structure + equivariance (~25 hrs)

Ties geometric DL, foundation models, biology, and quantum chemistry together.

- **AlphaFold 2** — AIQuraishi explainer (https://www.youtube.com/results?search_query=AlphaFold+2+explained+AIQuraishi) · paper (<https://www.nature.com/articles/s41586-021-03819-2>) · ESM/ESMFold · e3nn/EGNN

6.3 Integration project (~30 hrs) — your portfolio centerpiece

Something touching multiple domains — e.g., predict a molecular property with a hybrid quantum-classical model pretrained on a chemical foundation model, deployed as a monitored service. This is the artifact that demonstrates senior-level range.

6.4 Mock interviews + portfolio + writing (~35 hrs)

- **Recorded, reviewed:** 60-min ML systems design x several; 45-min segmentation deep-dive (defend loss/architecture choices); 45-min quantum (defend VQE/ADAPT/DMET to a skeptical chemist); 30-min behavioral (STAR, 4 stories prepped).
- **Read 1 paper/day** from your subfields; **write 2 short blog/LinkedIn posts** distilling something learned — writing for an audience exposes shaky understanding.

Cross-cutting habits (every week, no exceptions)

1. **Paper-of-the-week** — one paper, deeply read, derivations redone on paper. Don't break the streak.
 2. **Whiteboard Friday** — one concept whiteboarded without notes; photograph it. These become your interview-prep deck.
 3. **Re-implementation Saturday** — one concept from scratch in <200 lines. Repo: [weekly-rebuild/wk01-linalg](#), etc.
 4. **Sunday writeup** — a 1-page note explaining the week's deepest concept to a hypothetical junior. Teaching is the senior-level forcing function.
 5. **Senior-engineer "war chest"** — one Obsidian/Notion doc collecting derivations, paper one-liners, gotchas, anti-patterns. It becomes your interview safety net.
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Resource index (quick lookup)

Foundations math — 3B1B LA https://www.youtube.com/playlist?list=PLZHQObOWTQDPD3MizzM2xVFItgF8hE_ab · 3B1B Calc <https://www.youtube.com/playlist?list=PLZHQObOWTQDMsr9K-rj53DwVRMYO3t5Yr> · MIT 18.06 <https://ocw.mit.edu/courses/18-06-linear-algebra-spring-2010/> · MIT 18.065 <https://ocw.mit.edu/courses/18-065-matrix-methods-in-data-analysis-signal-processing-and-machine-learning-spring-2018/> · Stat 110 <https://projects.iq.harvard.edu/stat110/youtube> · MIT 6.041 <https://ocw.mit.edu/courses/res-6-012-introduction-to-probability-spring-2018/> · Matrix Cookbook <https://www.math.uwaterloo.ca/~hwolkowi/matrixcookbook.pdf> · Matrix Calculus <https://explained.ai/matrix-calculus/> · Karpathy Zero to Hero <https://www.youtube.com/playlist?list=PLAqhlrjkbWl23v9cThsA9GvCAUhRvKZ> · hub <https://karpathy.ai/zero-to-hero>

hero.html

Later-threaded math — Boyd <https://www.youtube.com/playlist?list=PL3940DD956CDF0622> · <https://web.stanford.edu/~boyd/cvxbook/> · Ruder <https://www.ruder.io/optimizing-gradient-descent/> · MacKay <https://www.inference.org.uk/itila/book.html> · Group theory (search) https://www.youtube.com/results?search_query=group+theory+for+physicists+lectures · GDL lectures <https://geometricdeeplearning.com/lectures/>

Medical imaging — MRI Q&A <https://mrquestions.com/index.html> · OHBM <https://www.youtube.com/@ohbmonline/playlists> · Callaghan https://www.youtube.com/playlist?list=PLqEMVgX2VgZcG_xR1QRdsuc-X_zEgMx-Z · iBiology <https://www.youtube.com/playlist?list=PLF513KEDjY9YBNhwMfX6jMI3PlpdW9TMP> · AI for Medical Diagnosis <https://www.coursera.org/learn/ai-for-medical-diagnosis> · DigitalSreeni <https://www.youtube.com/@DigitalSreeni/playlists> · code https://github.com/bnsreeni/python_for_microscopists · Conv arithmetic <https://arxiv.org/abs/1603.07285> · https://github.com/vdumoulin/conv_arithmetic · MONAI tutorials <https://github.com/Project-MONAI/tutorials> · MONAI YouTube <https://www.youtube.com/@projectmonai/videos> · nnU-Net <https://github.com/MIC-DKFZ/nnUNet> · TorchIO <https://torchio.readthedocs.io/> · Stanford AIMI <https://aimi.stanford.edu/education/educational-resources> · CS231n <https://www.youtube.com/playlist?list=PL3FW7Lu3i5jvHM8ljYj-zLfQRF3EO8sYv> · <https://cs231n.github.io/> · Metrics Reloaded <https://www.nature.com/articles/s41592-023-02151-z> *Segmentation papers*: U-Net <https://arxiv.org/abs/1505.04597> · V-Net <https://arxiv.org/abs/1606.04797> · Attention U-Net <https://arxiv.org/abs/1804.03999> · nnU-Net <https://www.nature.com/articles/s41592-020-01008-z> · Loss survey <https://arxiv.org/abs/2006.14822> · Generalized Dice <https://arxiv.org/abs/1707.03237> · Focal Tversky <https://arxiv.org/abs/1810.07842> · Boundary loss <https://arxiv.org/abs/1812.07032> · ViT <https://arxiv.org/abs/2010.11929> · TransUNet <https://arxiv.org/abs/2102.04306> · Swin-UNet <https://arxiv.org/abs/2105.05537> · UNETR <https://arxiv.org/abs/2103.10504> · Swin UNETR <https://arxiv.org/abs/2201.01266> · SAM <https://arxiv.org/abs/2304.02643> · MedSAM <https://www.nature.com/articles/s41467-024-44824-z> · MedNeXt <https://arxiv.org/abs/2303.09975>

Deep learning core — CS25 <https://web.stanford.edu/class/cs25/> · https://www.youtube.com/playlist?list=PLoROMvodv4rNijRchCzutFw5ltR_Z27CM · CS336 <https://stanford-cs336.github.io/spring2024/> · https://www.youtube.com/playlist?list=PLoROMvodv4rOY23Y0BoGoBGgQ1zmU_MT_ · CS236 <https://www.youtube.com/playlist?list=PLoROMvodv4rPOWA-omMM6STXaWW4FvjT8> · Lilian Weng <https://lilianweng.github.io/> · Annotated Transformer <http://nlp.seas.harvard.edu/annotated-transformer/> · HF NLP <https://huggingface.co/learn/nlp-course> · HF CV <https://huggingface.co/learn/computer-vision-course> · HF Diffusion <https://huggingface.co/learn/diffusion-course> *Papers*: RoPE <https://arxiv.org/abs/2104.09864> · RMSNorm <https://arxiv.org/abs/1910.07467> · FlashAttention <https://arxiv.org/abs/2205.14135> · GQA <https://arxiv.org/abs/2305.13245> · SwiGLU <https://arxiv.org/abs/2002.05202> · Switch/MoE <https://arxiv.org/abs/2101.03961> · ConvNeXt <https://arxiv.org/abs/2201.03545> · DINOv2 <https://arxiv.org/abs/2304.07193> · MAE <https://arxiv.org/abs/2111.06377> · DDPM <https://arxiv.org/abs/2006.11239> · Score-based (blog) <https://yang-song.net/blog/2021/score/> · Diffusion (blog) <https://lilianweng.github.io/posts/2021-07-11-diffusion-models/>

Quantum — IBM Quantum Learning <https://learning.quantum.ibm.com/> · Watrous YouTube <https://www.youtube.com/playlist?list=PLOFEBzvs-VvrXTMy5Y2lqmSaUjfnhvbHR> · Basics <https://learning.quantum.ibm.com/course/basics-of-quantum-information> · Algorithms <https://learning.quantum.ibm.com/course/fundamentals-of-quantum-algorithms> · General Formulation <https://learning.quantum.ibm.com/course/general-formulation-of-quantum-information> · Error Correction <https://learning.quantum.ibm.com/course/foundations-of-quantum-error-correction> · Qiskit YouTube <https://www.youtube.com/@qiskit/playlists> · QDA blog <https://www.ibm.com/quantum/blog/iql-migration> · PennyLane codebook <https://pennylane.ai/codebook> · PennyLane QML topic <https://pennylane.ai/topics/quantum-machine-learning> · PennyLane chemistry https://pennylane.ai/qml/demos_quantum-chemistry · MIT 8.04 <https://ocw.mit.edu/courses/8-04-quantum-physics-i-spring-2013/> · Quantum Country <https://quantum.country/qcvc> · Schuller https://www.youtube.com/playlist?list=PLPH7f_7ZlzxTi6kS4vCmv4ZKm9u8g5yic · Péré <https://github.com/Christophe-pere/QML-Course> · Hjorth-Jensen <https://github.com/CompPhysics/QuantumComputingMachineLearning> *Papers*: VQE <https://www.nature.com/articles/ncomms5213> · ADAPT-VQE <https://www.nature.com/articles/s41467-019-10988-2> · DMET-VQE <https://arxiv.org/abs/2108.08987> · QAOA <https://arxiv.org/abs/1411.4028> · Hadfield <https://arxiv.org/abs/1709.03489> · QML=kernels <https://arxiv.org/abs/2101.11020> · Barren plateaus <https://www.nature.com/articles/s41467-018-07090-4> · Expressibility <https://arxiv.org/abs/1905.10876> · Advantage-goal <https://arxiv.org/abs/2203.01340> · PQC encodings <https://arxiv.org/abs/2008.08605> · Pauli grouping <https://arxiv.org/abs/1907.13117>

Production / MLOps — CS329S <https://stanford-cs329s.github.io/> · Made With ML <https://madewithml.com/> · Full Stack DL <https://fullstackdeeplearning.com/course/2022/> · Chip Huyen <https://huyenchip.com/ml-interviews-book/> · Hamel Husain <https://www.oreilly.com/radar/what-we-learned-from-a-year-of-building-with-llms-part-i/> · Eugene Yan patterns <https://eugeneyan.com/writing/llm-patterns/> · Eugene Yan evals <https://eugeneyan.com/writing/llm-evaluators/> · Anthropic contextual retrieval <https://www.anthropic.com/news/contextual-retrieval> · Anthropic Constitutional AI <https://www.anthropic.com/news/constitutional-ai> · LlamaIndex <https://docs.llamaindex.ai/> · RAGAS <https://docs.ragas.io/> · DSPy <https://ds.py.ai/> · Raschka <https://magazine.sebastianraschka.com/> · HF multi-GPU https://huggingface.co/docs/transformers/en/perf_train_gpu_many · vLLM <https://arxiv.org/abs/2309.06180> · ZeRO <https://arxiv.org/abs/1910.02054>

Drug discovery / Geometric DL — Bronstein GDL <https://geometricdeeplearning.com/lectures/> · proto-book <https://arxiv.org/abs/2104.13478> · DeepChem <https://deepchem.io/tutorials/> · TDC <https://tdcommons.ai/> · SchNet <https://arxiv.org/a>

A note on the discomfort you described

Re-deriving things you already *use* feels like going backwards — and that feeling is the correct one for this transition, for the first stretch of Phase 1. It isn't backwards. Foundations-first is slower up front and pays compounding interest forever after: by the end of Phase 1 you'll read papers differently; by the end of Phase 4 you'll defend your own quantum work to a skeptical chemist without flinching. There's no deadline now, so let each phase take the time it needs, pass its self-test before moving on, and protect your rest. This is a scaffold for durable mastery, not a race.